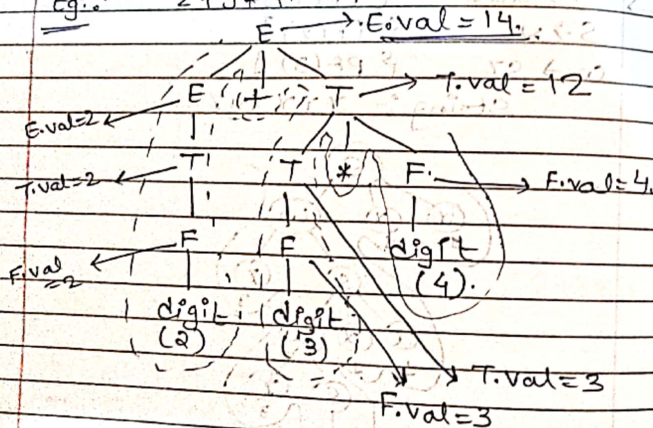


* SYNTAX DIRECTED DEFINITION (SDD):

SDD = Grammar + Semantic Rules.

<u>Production</u>	<u>Semantic Rules</u>
$E \rightarrow E + T$	$E.val = E.val + T.val$
$E \rightarrow T$	$E.val = T.val$
$T \rightarrow T * F$	$T.val = T.val * F.val$
$T \rightarrow F$	$T.val = F.val$
$F \rightarrow \text{digit}$	$F.val = \text{digit.val}$ (lexical.)

Eg.:- $2 + 3 * 4$



Q. Prod?

<u>Prod.ⁿ</u>	<u>Semantic Rule</u>
$E \rightarrow E \# T$	$E.val = E.val \# T.val$
$E \rightarrow T$	$E.val = T.val$
$T \rightarrow T \& F$	$T.val =$
$T \rightarrow F$	$T.val = F.val$
$F \rightarrow digit$	$F.val = digit.val$

(3) Expr:- $8 \# 12 \& 4 \# 16 \& 12 \# 4 \& 2 = 512$
 $\therefore T.val = (?)$

- ☐ A $T.val = T.val * F.val$
☐ B $T.val = T.val + F.val$
☒ C $T.val = T.val - F.val$
☐ D NOT.

$$8 \# 12 \& 4 \# 16 \& 12 \# 4 \& 2.$$

$$8 * (12 + 4) * (16 + 12) * (4 + 2)$$

$$= 8 * 16 * 28 * 6 \neq 512$$

$$8 * (12 - 4) * (16 - 12) * (4 - 2) = 8 * 8 * 4 * 2 = 512$$

2. $10 \# 8 \& 6 \# 9 \& 4 \# 5 \& 2 = (?)$

$$= 10 * (8 - 6) * (9 - 4) * (5 - 2)$$
$$= 10 * 2 * 5 * 3 = 300.$$

⇒ Types of SDD

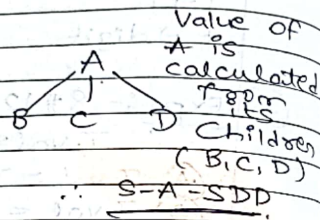
- (1) S-Attributed :- A SDD that uses only synthesized attribute is called S-Attributed SDD.

Eg. :- $A \rightarrow BCD$

$A.i = B.i$

$A.i = C.i$

$A.i = D.i$



→ Semantic actions are always placed at right end of the prod.
[Postfix SDD].

→ Attributes are evaluated with BUP. (Bottom Up Parser).

- (2) L-Attributed :- A SDD that uses both synthesized & inherited attribute is called as L-attributed SDD. But, each inherited attribute is restricted to inherit from Parent or left siblings only.

Eg. :- $A \rightarrow BCD$

$C.i = A.i$

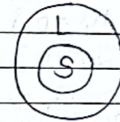
$C.i = B.i$

$C.i = D.i$ X (∵ Calculated from Right sibling)

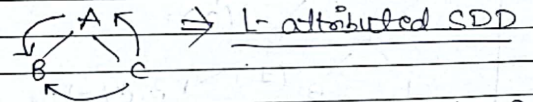
→ Semantic actions are placed anywhere on R.H.S. of the prod.

→ Attributes are evaluated by traversing Parse tree depth first, left to right.

⇒ Relation :-

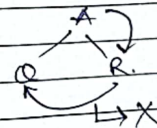


① $A \rightarrow BC$ { $B.i = A.i$, $C.i = B.i$, $A.i = C.i$ }



⇒ L-attributed SDD

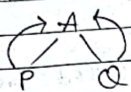
② $A \rightarrow QR$ { $R.i = A.i$, $Q.i = R.i$, $A.i = Q.i$ }



L-A-SDD X

S-A-SDD X

Q. $A \rightarrow PQ$ { $A.i = P.i$, $A.i = Q.i$ }.



S-attributed SDD...

SDD to Convert Binary to Decimal.

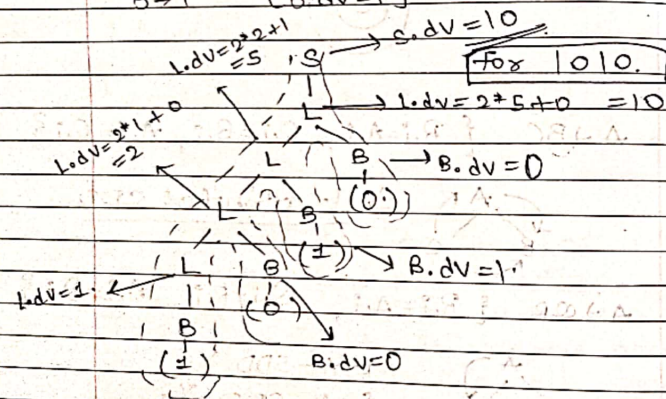
$S \rightarrow L$ { $S.dv = L.dv$ }

$L \rightarrow LB$ { $L.dv = 2 * L.dv + B.dv$ }

$L \rightarrow B$ { $L.dv = B.dv$ }

$B \rightarrow 0$ { $B.dv = 0$ }

$B \rightarrow 1$ { $B.dv = 1$ }



Binary to Decimal with fraction.

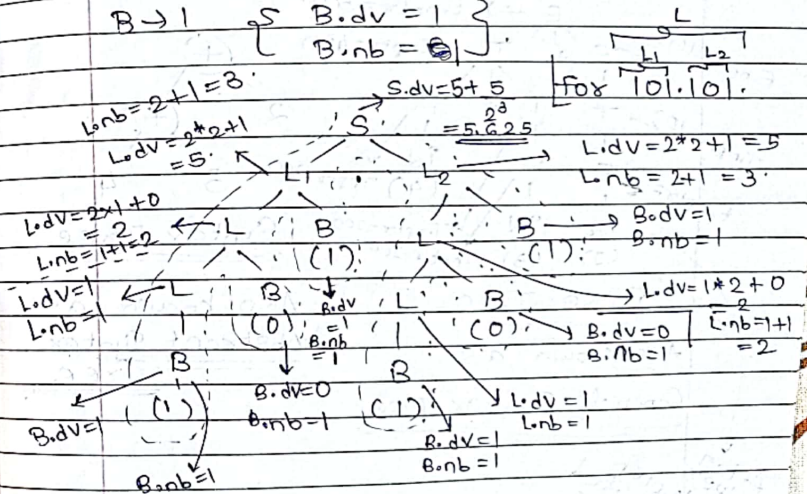
$S \rightarrow L_1 L_2$ { $S.dv = L_1.dv + \frac{L_2.dv}{2^{L_2.nb}}$ } No. of bits.

$L \rightarrow LB$ { $L.dv = 2 * L.dv + B.dv$ }
 $L.nb = L.nb + B.nb$

$L \rightarrow B$ { $L.dv = B.dv$ }
 $L.nb = B.nb$

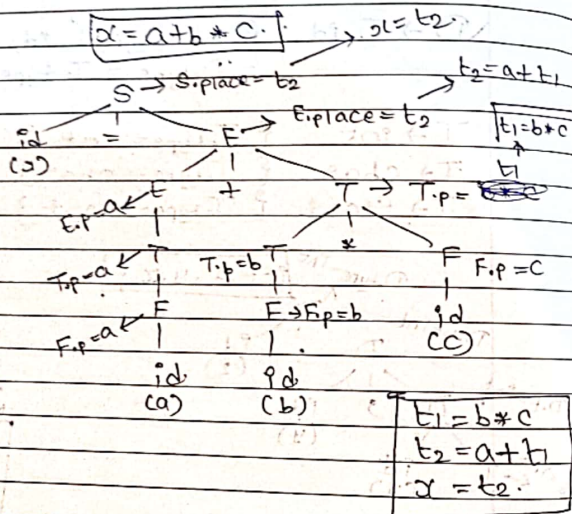
$B \rightarrow 0$ { $B.dv = 0$ }
 $B.nb = 1$

$B \rightarrow 1$ { $B.dv = 1$ }
 $B.nb = 1$



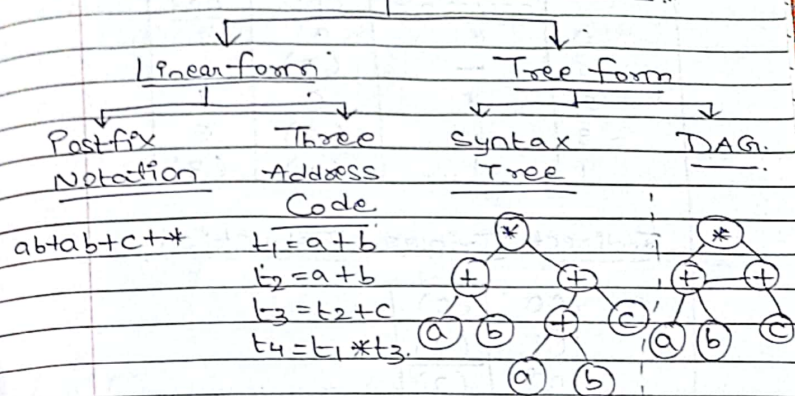
⇒ How to generate Three Address Code

$S \rightarrow id = E$ { gen(id.name = E.place) }
 $E \rightarrow E + T$ { E.place = new temp();
 gen(E.place = E.place + T.place) }
 $E \rightarrow T$ { E.place = T.place }
 $T \rightarrow T * F$ { T.place = new temp();
 gen(T.place = T.place * F.place) }
 $T \rightarrow F$ { T.place = F.place }
 $F \rightarrow id$ { F.place = id.name }



Translates High-Level source code into machine-independent simplified representation.

⇒ Intermediate Code Generation



⇒ Representation of Three Address Code

Example expression: $-(a * b) + (c * d + e)$

Quadruples					
	Operator	Op1	Op2	result	
$t_1 = a * b$	*	a	b	t1	
$t_2 = -t_1$	-	t1		t2	
$t_3 = c * d$	*	c	d	t3	
$t_4 = t_3 + e$	+	t3	e	t4	
$t_5 = t_2 + t_4$	+	t2	t4	t5	

Triples:

	operator	op1	op2
0	*	a	b
1	-	(0)	
2	*	c	d
3	+	(2)	e
4	+	(1)	(3)

Indirect Triples Instruction:

100	(0)
101	(1)
102	(2)
103	(3)
104	(4)

⇒ Directed Acyclic Graphs (DAG)

$$X = (((a+a) + (a+a)) + ((a+a) + (a+a)))$$

$$t_1 = a+a$$

$$t_2 = a+a$$

$$t_3 = a+a$$

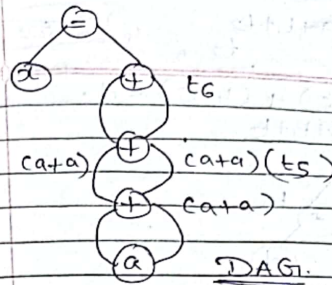
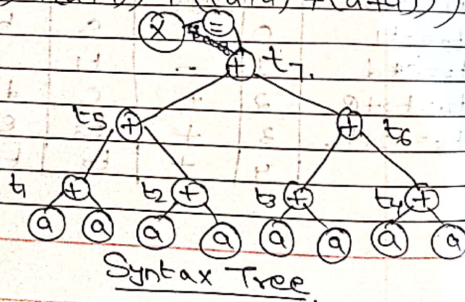
$$t_4 = a+a$$

$$t_5 = t_1 + t_2$$

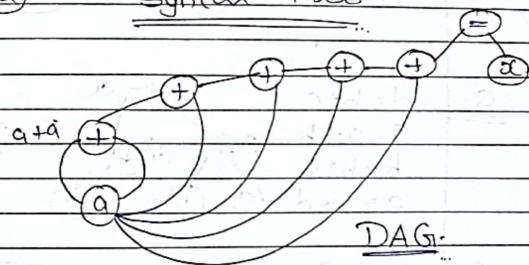
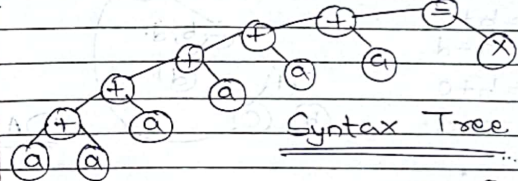
$$t_6 = t_3 + t_4$$

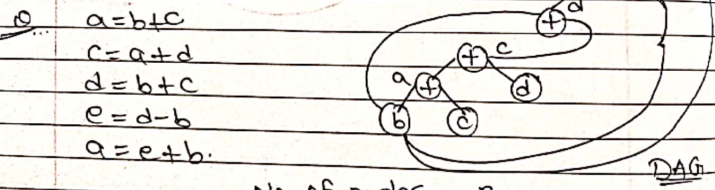
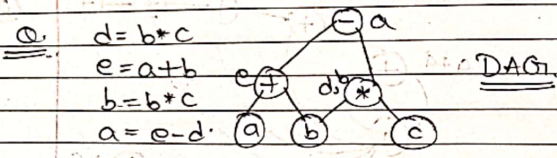
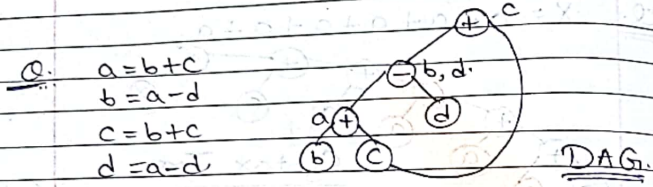
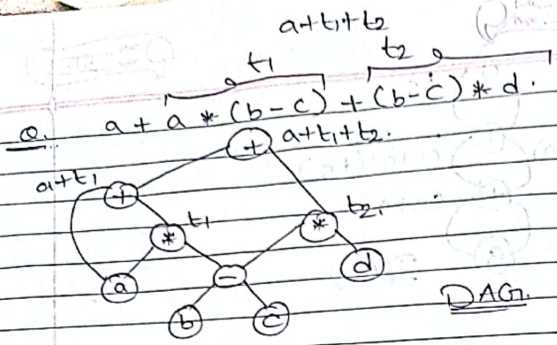
$$t_7 = t_5 + t_6$$

$$x = t_7$$



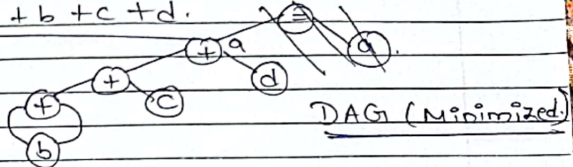
Q. $X = a + a + a + a + a + a$





No. of nodes = 8
 No. of edges = 10.

$a = e + b$
 $= d - b + b = d$
 $a = b + c$
 $a = b + a + d$
 $a = b + b + c + d$



No. of nodes = 6
 No. of edges = 6.